Yixuan Li

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Education

Southern University of Science and Technology (SUSTech) **Aug. 2022 – Jun. 2026 (Expected)**
Bachelor of Science in Data Science *Shenzhen, China*

- **GPA:** 3.79/4.00
- **Mathematics Courses:** Linear Algebra, Mathematical Analysis, Discrete Mathematics and Its Applications, Foundation of Probability Theory, Mathematical Statistics, Statistical Linear Models, Operational Research and Optimization
- **CS Courses:** Introduction to Computer Programming, Data Structures and Algorithm Analysis, Principles of Database Systems, Distributed Storage and Parallel Computing, Advanced Natural Language Processing, Artificial Intelligence

University of California, San Diego (UCSD) **Mar. 2025 – Jun. 2025**
University and Professional Studies (UPS) Program *San Diego, CA, United States*

- **GPA:** 4.00/4.00
- **Courses:** Machine Learning Algorithms, AI Search and Reasoning, Topics of Data Science: Trustworthy Machine Learning

Research Interest

- **Trustworthy Machine Learning**: Uncertainty Quantification, Hallucination Detection, Model Interpretability, LLM Alignment, Generative AI
- **AI for Science (Biomedical)**: Multimodal Learning, Graph Neural Networks (GNNs), Medical Imaging

Research Experience

SUSTech Machine Learning Lab | *Advisor: Prof. Hongxin Wei* | SUSTech **Shenzhen, China**

Project: Multimodal Learning for Spatial Transcriptomics-to-Proteomics Prediction
Main Developer *Oct. 2025 – Present*

- Built an end-to-end framework for the STP Open Challenge, translating spatial transcriptomics into proteomic distributions on adjacent glioma tissue sections.
- Developed *UNet* and later *GCN/GAT-based graph models*, introducing a patch-based neighbor-sampling strategy to mitigate boundary bias and improve generalization.
- Designed a *multimodal fusion pipeline* integrating ResNet-derived morphological features from H&E Image with RNA embeddings, boosting performance to **0.74** on Top 10 Average Spearman Correlation and ranking **Top 1** on the official leaderboard .
- Generated biologically coherent protein-level spatial maps that preserved tissue architecture and hotspot regions, enabling downstream analyses such as cellular microenvironment profiling.

Project: Benchmarking Uncertainty Quantification Methods in Large Language Models
Independent Researcher *Jun. 2025 – Present*

- Built a unified benchmarking pipeline for 10+ *semantic uncertainty quantification methods* (e.g., SE, DSE, SAR) on QA datasets using open-source LLMs, evaluated via AUROC, AUARC, and PRR.
- Standardized data processing and evaluation for reproducible comparison across uncertainty estimation approaches.
- Conducted an extensive literature review on *semantic uncertainty in LLMs*, connecting with hallucination detection and calibration, and published findings in technical blog posts.

Jianqing Shi's Lab | *Advisor: Prof.Jianqing Shi* | SUSTech **Shenzhen, China**

Project: Statistical Analysis of Mental Health Impact among Menopausal Women
Core Collaborator *Sep. 2025 – Nov. 2025*

- Analyzed large-scale survey data on menopausal women in Malaysia to examine the relationship between menopausal stages and mental health outcomes.
- Applied statistical methods including *Kruskal–Wallis, Fisher-Exact, Chi-square, ANCOVA, CMH, Paired t-tests*, and *multiple imputation* for hypothesis testing and subgroup analysis.
- Identified stage-adjusted psychological risk factors and enhanced inference robustness in women's health research.
- Due to data confidentiality, results are restricted to internal reports.

Internship Experience

Machine Learning for Clinical Decision Support in Orthodontics **Jun. 2024 – Aug. 2024**
Summer Internship — Mentor: Dr. Ying Shan *Peking University Shenzhen Hospital*

- Performed data preprocessing on clinical craniofacial imaging datasets, dental records, and treatment histories, which include cleaning, de-duplication, and standardization.
- Developed *linear and neural network models* based on craniofacial measurement variables to predict patient-specific tooth extraction plans. Achieved promising prediction accuracy and facilitated internal adoption.

Personal Projects & Competitions

Concept Bottleneck in Generative Models: Reproduction Study | *Core Member* **Apr. 2025 – Jun. 2025**

- Reproduced the CVPR paper on Post-hoc Concept Bottleneck frameworks: *Concept Bottleneck Autoencoder (CB-AE)*, *Concept Controller (CC)* using *StyleGAN*, replicating key quantitative and qualitative results from the original study.
- Conducted additional experiments on *Concept Extra-Interpolation, Entanglement, and Leakage* to evaluate model interpretability, robustness, and reliability under concept-level perturbations.
- Delivered a comprehensive reproduction package including *Github code repository*, *presentation slides*, and a public *technical blog*, achieving a course score of **43.4/40 (Top 1)** .

Kaggle Competition: Fine-Tuning Gemma-2B for Chinese QA Dataset | *Team Leader* **Oct. 2024 – Jan. 2025**

- Led a team to fine-tune *Gemma-2B* on the *GPT-4 Chinese QA dataset*. Built a complete fine-tuning pipeline covering data preprocessing, LoRA integration, and evaluation. Published notebook can be viewed *here*.
- Integrated *Retrieval-Augmented Generation (RAG)* for data augmentation and *LoRA-based PEFT* for parameter-efficient fine-tuning, resulting in improved dataset diversity, contextual grounding, and instruction-following performance.
- Evaluated model performance using *BLEU, ROUGE, METEOR, and BERTScore metrics*, and performed *cross-lingual robustness analysis* showing stable results across four languages.

Awards

- **First-class Scholarship for Outstanding students, SUSTech** (Top 5%) **2023 – 2024, 2024 – 2025**
- **Third-class Scholarship for Outstanding students, SUSTech** (Top 15%) **2022 – 2023**
- **Second Prize** Awarded in the 2024 China Undergraduate Mathematical Contest in Modeling (CUMCM), in Guangdong division **Oct. 2024**
- **Honorable Mention** Awarded in the 2024 Mathematical Contest In Modeling (MCM) **May 2024**
- **Excellent Student Award** For outstanding students in SUSTech for three times **2022 – 2025**

Leadership / Extracurricular

Community Involvement **Oct. 2023 – Present**
Volunteer *SUSTech*

- Participated in online teaching programs, delivering educational lessons to children in rural areas.
- Accumulated over **179 hours** of volunteer service over three years.
- Recognized as an **Outstanding Volunteer** at both the university and School of Science levels.

Media of Shuren College, SUSTech **Sep. 2023 – Aug. 2024**
President *SUSTech*

- Leading a team with 15 Team members. Created numerous contents/posts for the college's official media account.
- Received the **Excellent College Student Leader Award** and the **Outstanding College Organizations Award**.

Pickleball Team **Oct. 2023 – Present**
Member *SUSTech*

- Represented SUSTech in the Second Guangdong Pickleball Competition (Women's Doubles, **Top 16** finish)
- Participated in the 2023 SUSTech Pickleball Championship, achieving **5th place** in Mixed Doubles.

Skills

- **Programming:** Python, R, SQL, Java, JavaScript, HTML, Hadoop
- **Development Tools:** PyTorch, Git, Jupyter Notebook, LaTeX, Markdown
- **Languages:** English (Fluent, TOEFL 109, GRE 322), Mandarin (Native), Cantonese (Fluent)